Hints: Differentiable functions – 1

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Decompose into partial fractions before differentiating: $\dfrac{x}{x^2-1} = \dfrac{1}{2}\Big(\dfrac{1}{x-1} + \dfrac{1}{x+1}\Big)$.

Hint 2. Use $\big(\frac{1}{x-a}\big)^{(n)} = \frac{(-1)^n n!}{(x-a)^{n+1}}$. The answer is $f^{(n)}(x) = \dfrac{(-1)^n n!}{2}\Big(\dfrac{1}{(x-1)^{n+1}} + \dfrac{1}{(x+1)^{n+1}}\Big)$.

Problem 2

Hint 1. Since $f^{(k)}(0) = k!\,a_k$, where $f = \sum a_k x^k$ near 0, you have to show that all Taylor coefficients of f are positive. Products are awkward, so take logarithms.

Hint 2. $\ln f(x) = -\sum_i \ln(1 - c_i x) = \sum_{k\geqslant 1} \dfrac{s_k}{k} x^k$ with $s_k = c_1^k + \cdots + c_n^k > 0$. So $f = e^g$, where g has zero constant term and positive coefficients.

Hint 3. From $f' = g'f$, compare coefficients: $a_0 = 1$ and $n a_n = \sum_{k=1}^{n} s_k\, a_{n-k}$. Induction on n gives $a_n > 0$.

Problem 3

Hint 1. The hypothesis gives information about $(f^2)' = 2ff'$. Suppose that $f(x) \to L$ and look for a contradiction.

Hint 2. Integrate $(f^2)' \geqslant 2\cos x$ over an interval where $\cos x \geqslant 0$, e.g. $[2k\pi - \frac{\pi}{2},\, 2k\pi + \frac{\pi}{2}]$. This gives $f^2(2k\pi + \frac{\pi}{2}) - f^2(2k\pi - \frac{\pi}{2}) \geqslant 4$ for every k.

Hint 3. If $f \to L$, then $f^2 \to L^2$, so these differences must tend to 0. Contradiction. (Boundedness of f rules out an infinite limit, so only finite L has to be considered.)

Problem 4

Hint 1. This is the trick of Theorem 2 with a different multiplier. Find $\varphi > 0$ such that $(\varphi f)' = x\varphi\,(f + \frac{f'}{x})$.

Hint 2. $\varphi = e^{x^2/2}$ works. Write $f = \dfrac{e^{x^2/2} f}{e^{x^2/2}}$ and apply l'Hôpital (the denominator tends to ∞, which is all l'Hôpital needs).

Hint 3. The ratio of derivatives is $\dfrac{(f' + xf)e^{x^2/2}}{xe^{x^2/2}} = f + \dfrac{f'}{x} \to 0$, so $f \to 0$.

Problem 5

Hint 1. Write $\frac{\sin x}{x}$ as an integral with a parameter: $\dfrac{\sin x}{x} = \displaystyle\int_0^1 \cos(xt)\,dt$.

Hint 2. Differentiate under the integral sign: $f^{(n)}(x) = \displaystyle\int_0^1 t^n \cos\big(xt + \tfrac{n\pi}{2}\big)\,dt$.

Hint 3. Bound $|\cos| \leqslant 1$ to get $|f^{(n)}(x)| \leqslant \int_0^1 t^n\,dt = \frac{1}{n+1}$. The inequality is strict because the continuous integrand $|\cos(xt + n\pi/2)|$ is not identically 1 on $[0,1]$ for $x > 0$.